Propositional Logic and Propositional Equivalences

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Propositional Logic

Definition of Proposition

Proposition

A **proposition** is a declarative sentence (a sentence that declares a fact) that is either **true** or **false**, but not both.

- Propositions are usually denoted by **letters**, such as $p, q, r, \ldots$
- Example:
  - p: Washington, D.C. is the capital of the USA.
  - q: Toronto is the capital of Canada.

Examples of Valid Propositions

1. Washington, D.C. is the capital of the United States of America. (True)
2. $1 + 1 = 2$. (True)
3. $2 + 2 = 3$. (False)

Each of these is a declarative sentence that is either true or false.

Examples that are NOT Propositions

1. What time is it? (Question, not declarative)
2. Read this carefully. (Command, not declarative)
3. $x + 1 = 2$. (Open sentence, depends on x)
4. $x + y = z$. (Open sentence, depends on variables)

These are not propositions because they are neither true nor false without additional information.

Logical Connectives

- Connectives combine propositions to form **compound propositions**.
- Examples of connectives:
  - Negation ($\neg$)
  - Conjunction ($\wedge$)
  - Disjunction ($\vee$)
  - Exclusive OR (XOR, $\oplus$)
  - Implication ($\rightarrow$)
  - Bi-conditional ($\leftrightarrow$)

Negation of a Proposition

Definition

Let p be a proposition. The **negation** of p, denoted by $\neg p$ (or $\bar{p}$), is the statement:

"It is not the case that p."

The truth value of $\neg p$ is the opposite of the truth value of p.

Example

Proposition: "Michael's PC runs Linux."
Negation: "It is not the case that Michael's PC runs Linux."
Simplified: "Michael's PC does not run Linux."

Truth Table of Negation

p	$\neg p$
T	F
F	T

Table: Truth Table for Negation

Conjunction of Propositions

Definition

Let p and q be propositions. The **conjunction** of p and q, denoted by $p \wedge q$, is the proposition

"p and q."

The conjunction $p \wedge q$ is true when both p and q are true, and false otherwise.

Example

p: "Rebecca's PC has more than 16 GB free hard disk space."
q: "The processor in Rebecca's PC runs faster than 1 GHz."

Conjunction: "Rebecca's PC has more than 16 GB free hard disk space, **and** the processor in Rebecca's PC runs faster than 1 GHz."

Truth Table for Conjunction

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

The conjunction is true only when both p and q are true. Otherwise it is false.

Disjunction of Propositions

Definition

Let p and q be propositions. The **disjunction** of p and q, denoted by $p \vee q$, is the proposition

"p or q."

The disjunction $p \vee q$ is true when at least one of p or q is true, and false only when both p and q are false.

Example

p: "It is raining."
q: "It is windy."

Disjunction: "It is raining **or** it is windy (or both)."

Truth Table for Disjunction

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

The disjunction is false only when both p and q are false. This statement is true if either (or both) conditions are true.

Exclusive OR (XOR)

Definition

Let p and q be propositions. The **exclusive or** of p and q, denoted by $p \oplus q$, is the proposition that is true when **exactly one** of p and q is true, and false otherwise.

Example

p: "The light is on."
q: "The fan is on."

Exclusive OR: "Either the light is on **or** the fan is on, but not both."

Truth Table for Exclusive OR

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

The exclusive OR is true only when exactly one of p or q is true. It is false if both are on or both are off.

Conditional Statement (Implication)

Definition

Let p and q be propositions. The **conditional statement** $p \rightarrow q$ is the proposition

"If p, then q."

It is false only when p is true and q is false; otherwise it is true. p is the *hypothesis* (antecedent/premise) and q is the *conclusion* (consequence).

Example

p: "Rebecca studies hard."
q: "Rebecca will pass the exam."
Conditional statement: "If Rebecca studies hard, then she will pass the exam."

Conditional Statement (Implication)

Alternate Terminology

The statement $p \rightarrow q$ can be expressed in many ways:

- "If p, then q"
- "p implies q"
- "p is sufficient for q"
- "q whenever p"
- "A necessary condition for p is q"
- "q follows from p"

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	T	T

The conditional $p \rightarrow q$ is false only when p is true and q is false. In all other cases it is true.

Converse, Contrapositive, and Inverse

Definitions

Let $p \to q$ be a conditional statement.

- **Converse:** $q \to p$
- **Contrapositive:** $\neg q \to \neg p$
- **Inverse:** $\neg p \to \neg q$

Note: Only the contrapositive is logically equivalent to the original conditional $p \to q$.

Example: Converse, Contrapositive, and Inverse

Original statement: *"If it is raining, then the home team wins."*

- **Contrapositive:** "If the home team does not win, then it is not raining."
- **Converse:** "If the home team wins, then it is raining."
- **Inverse:** "If it is not raining, then the home team does not win."

Only the contrapositive is equivalent to the original statement.

Biconditional Statement

Definition

Let p and q be propositions. The **biconditional statement** $p \leftrightarrow q$ is the proposition

"p if and only if q."

It is true when p and q have the same truth value, and false otherwise.

Biconditional statements are also called *bi-implications*.

Alternate Terminology

$p \leftrightarrow q$ can also be expressed as:

- "p is necessary and sufficient for q"
- "If p, then q, and conversely"
- "p iff q"

Truth Table for Biconditional

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

The biconditional $p \leftrightarrow q$ is true when both $p \rightarrow q$ and $q \rightarrow p$ are true. In other words, $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$.

Example

p: "You can enter the lab."
q: "You have an ID card."
Biconditional: "You can enter the lab if and only if you have an ID card."

Example: Compound Proposition

Construct the truth table of the compound proposition:

$$(p \vee \neg q) \rightarrow (p \wedge q)$$

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

The table is built step by step: first compute $\neg q$, then $p \vee \neg q$, then $p \wedge q$, and finally the compound proposition.

Introduction to Bitwise Operations

Bits and Truth Values

- A **bit** is a symbol with two values: 0 (false) and 1 (true).
- A **Boolean variable** has value 1 if true, 0 if false.
- Computers represent information using bits.

Bitwise Operations

- Bitwise operations correspond to logical connectives.
- Replace T (true) with 1 and F (false) with 0 in truth tables.
- Common operators: OR ($\vee$), AND ($\wedge$), XOR ($\oplus$).

Truth Table for Bitwise Operations

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Bitwise operators act on each pair of bits independently.

Example: Bitwise OR, AND, XOR

Given bit strings:

$$01\,1011\,0110 \quad \text{and} \quad 11\,0001\,1101$$

- **Bitwise OR:** $11\,1011\,1111$
- **Bitwise AND:** $01\,0001\,0100$
- **Bitwise XOR:** $10\,1010\,1011$

Each result is computed by applying OR, AND, XOR to corresponding bits.

Example: Translate to Logic

English

"The automated reply cannot be sent when the file system is full."

Propositional Variables

- p: "The automated reply can be sent."
- q: "The file system is full."

Translation

When $=$ "if" $\Rightarrow$ $q \rightarrow \neg p$

Reasoning: "cannot be sent when q" means "if q then not p."

Example: Translate to Logic

English

"You cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old."

Propositional Variables

- q: "You can ride the roller coaster."
- r: "You are under 4 feet tall."
- s: "You are older than 16 years old."

Translation

unless $s \equiv$ "if not s" $\Rightarrow \quad (r \wedge \neg s) \rightarrow \neg q$

If you are short and not at least 16, then you cannot ride.

Example: Translate to Logic

English

"You can access the Internet from campus only if you are a computer science major or you are not a freshman."

Propositional Variables

- a: "You can access the Internet from campus."
- c: "You are a computer science major."
- f: "You are a freshman."

Translation

only if $\Rightarrow$ necessary condition $\quad a \rightarrow (c \vee \neg f)$

Access implies (CS major) or (not a freshman).

Propositional Equivalences

Tautology, Contradiction, Contingency

Definitions

- **Tautology:** A compound proposition that is *always true*, regardless of the truth values of its variables.
- **Contradiction:** A compound proposition that is *always false*.
- **Contingency:** A compound proposition that is *sometimes true and sometimes false*.

Examples: $p \vee \neg p$ is a tautology; $p \wedge \neg p$ is a contradiction; $p \rightarrow q$ is a contingency.

Propositional Equivalence: Definition, Notation, Example

Definition & Notation

Two compound propositions p and q are **logically equivalent** if $p \leftrightarrow q$ is a *tautology*.
We write $p \equiv q$.

Example (statement vs. contrapositive form)

$p \rightarrow q \equiv \neg p \vee q$ (they have identical truth values)

Equivalence can be shown either by a **truth table** or by using the **equivalence laws** below.

Logical Equivalence Laws (1/2)

Identity

$p \wedge T \equiv p$

$p \vee F \equiv p$

Idempotent

$p \vee p \equiv p$

$p \wedge p \equiv p$

Domination

$p \vee T \equiv T$

$p \wedge F \equiv F$

Commutative

$p \vee q \equiv q \vee p$

$p \wedge q \equiv q \wedge p$

Double Negation

$\neg(\neg p) \equiv p$

Logical Equivalence Laws (2/2)

Associative

$(p \vee q) \vee r \equiv p \vee (q \vee r)$
$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$

Absorption

$p \vee (p \wedge q) \equiv p$
$p \wedge (p \vee q) \equiv p$

Distributive

$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

Negation

$p \vee \neg p \equiv T$
$p \wedge \neg p \equiv F$

De Morgan

$\neg(p \wedge q) \equiv \neg p \vee \neg q$
$\neg(p \vee q) \equiv \neg p \wedge \neg q$

Implication

$p \rightarrow q \equiv \neg p \vee q$

Example (Truth Table): $\neg p \vee q \equiv p \rightarrow q$

Show that $\neg p \vee q \equiv p \rightarrow q$.

p	q	$\neg p \vee q$	$p \rightarrow q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

The columns associated with $\neg p \vee q$ and $p \rightarrow q$ match for all rows, hence $\neg p \vee q \equiv p \rightarrow q$.

Example (Truth Table): Distributive Law

Show that $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

p	q	r	$q \wedge r$	$p \vee q$	$p \vee r$	$p \vee (q \wedge r)$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	T	F	F	F
F	F	T	F	F	T	F	F
F	F	F	F	F	F	F	F

The columns associated with $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ match for all rows, hence $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.

Example (Using Laws)

Show that $\neg\big(p \vee (\neg p \wedge q)\big)$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences.

$$
\begin{aligned}
\neg\big(p \vee (\neg p \wedge q)\big) &\equiv \neg p \wedge \neg(\neg p \wedge q) && \text{De Morgan} \\
&\equiv \neg p \wedge \big(\neg\neg p \vee \neg q\big) && \text{De Morgan} \\
&\equiv \neg p \wedge (p \vee \neg q) && \text{Double Negation} \\
&\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && \text{Distributive} \\
&\equiv F \vee (\neg p \wedge \neg q) && \text{Negation } (p \wedge \neg p) \\
&\equiv \neg p \wedge \neg q && \text{Identity } (F)
\end{aligned}
$$

Example (Using Laws)

Show that $\neg p \to (q \to r)$ and $q \to (p \vee r)$ are logically equivalent by developing a series of logical equivalences.

$$
\begin{aligned}
\neg p \to (q \to r) &\equiv \neg(\neg p) \vee (q \to r) && \text{Implication law } (a \to b \equiv \neg a \vee b) \\
&\equiv p \vee (q \to r) && \text{Double Negation} \\
&\equiv p \vee (\neg q \vee r) && \text{Implication law} \\
&\equiv (p \vee \neg q) \vee r && \text{Associative} \\
&\equiv (\neg q \vee p) \vee r && \text{Commutative} \\
&\equiv \neg q \vee (p \vee r) && \text{Associative} \\
&\equiv q \to (p \vee r) && \text{Implication law (reverse)}
\end{aligned}
$$

Propositional Satisfiability

Definition

A compound proposition is **satisfiable** if there exists an assignment of truth values to its variables that makes it **true**.
A compound proposition is **unsatisfiable** if it is false for all possible assignments of truth values to its variables.

Note: A compound proposition is unsatisfiable $\iff$ its negation is a tautology.

Example: Propositional Satisfiability

Determine whether $(p \vee \neg q) \wedge (q \vee \neg r) \wedge (r \vee \neg p)$ is satisfiable.

Solution

Consider truth assignment: $p = T, q = T, r = T$.

- $p \vee \neg q = T \vee F = T$
- $q \vee \neg r = T \vee F = T$
- $r \vee \neg p = T \vee F = T$

Since all clauses evaluate to true, the compound proposition is **satisfiable**.

Therefore, the formula is satisfiable because at least one assignment makes it true.